Vision for Open: Building an Effective Control for the AI Data Center Era

The Missing Control Layer in the AI Data Center Stack

B Parthiban

CVP Boot FW

Munir Ahmad

Chief Compute Architect

OPEN Compute Project

2026 OCP GLOBAL SUMMIT

AI Scale, Management Becomes System Control

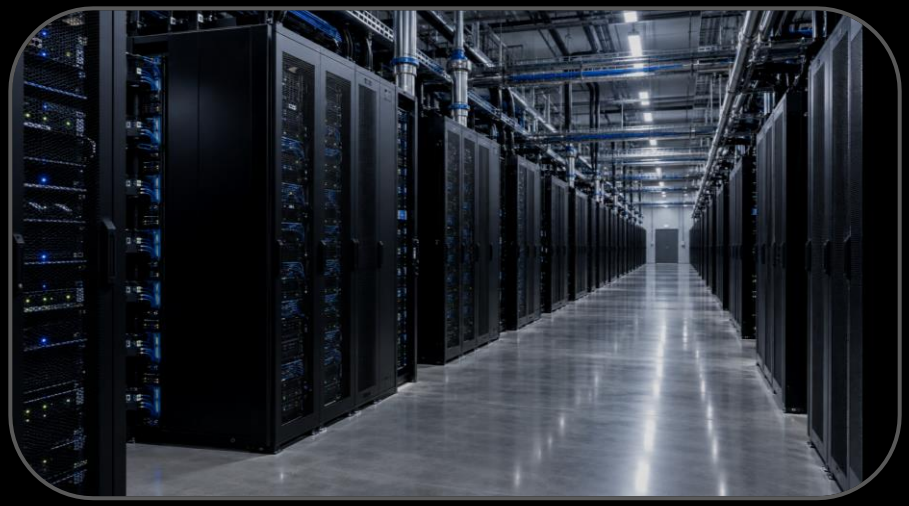
SERVER-CENTRIC



Manage the server

BMC • basic server lifecycle • device-level control

SYSTEM SCALE



Control the system

Compute • power • cooling • network • lifecycle • security • telemetry

System control coordinates the functions that determine how the rack operates, responds and recovers

The AI data center is becoming a system of systems

01 DEVICE TRUST STOPS TOO EARLY
Attestation often terminates at the component boundary.

02 PLATFORM VARIATION MULTIPLIES
Control and telemetry differ across platforms, ODMs and generations.

03 TOPOLOGY IS NOT VERIFIED
As-built composition is visible, but not independently provable.

04 BRING-UP AND RECOVERY ARE PROPRIETARY
Authority, sequencing and failure behavior are left to implementation.



System Control Is a Cross-Layer Problem

OPERATORS
Run and deploy at scale

MANAGEMENT
Policy, lifecycle and fleet state

FIRMWARE
Initialize, secure and recover the platform

HARDWARE TRUST
Establish identity and integrity

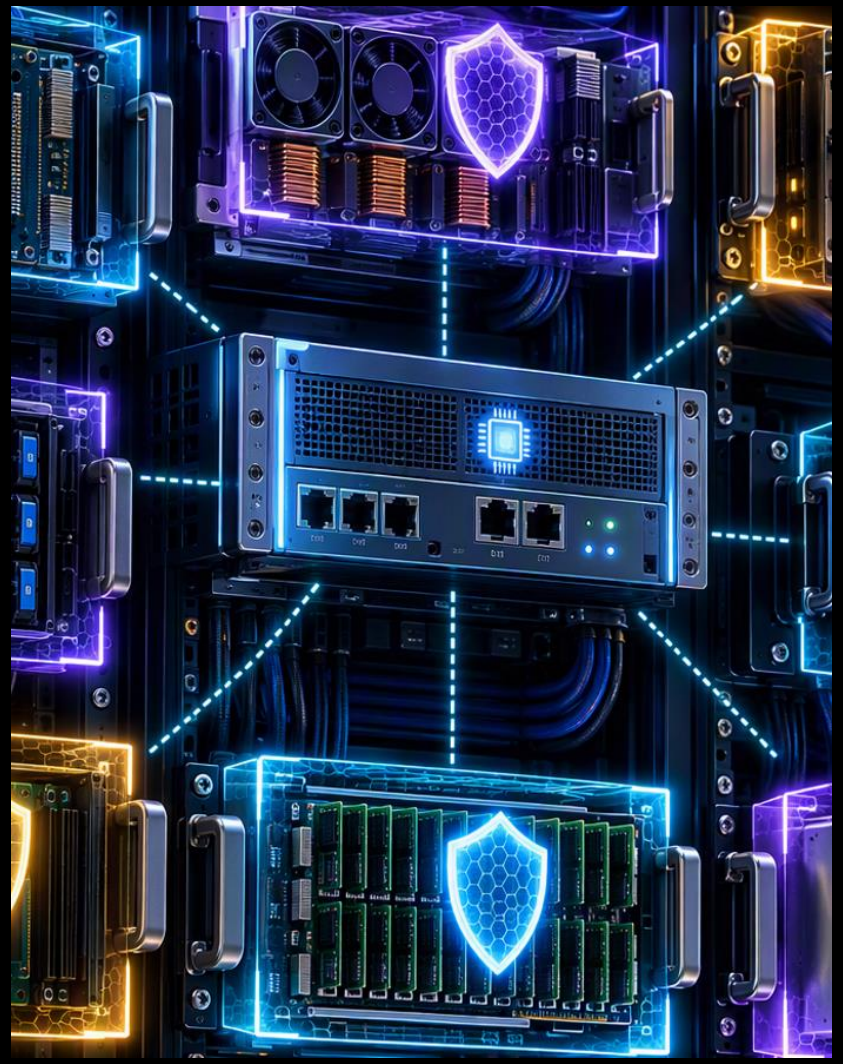
SILICON
Compute, manage and accelerate

IMPLICATION

If the layers operate independently, system-level control breaks down

The architecture must coordinate state, trust and action across the full stack.

Management interfaces alone do not create effective control



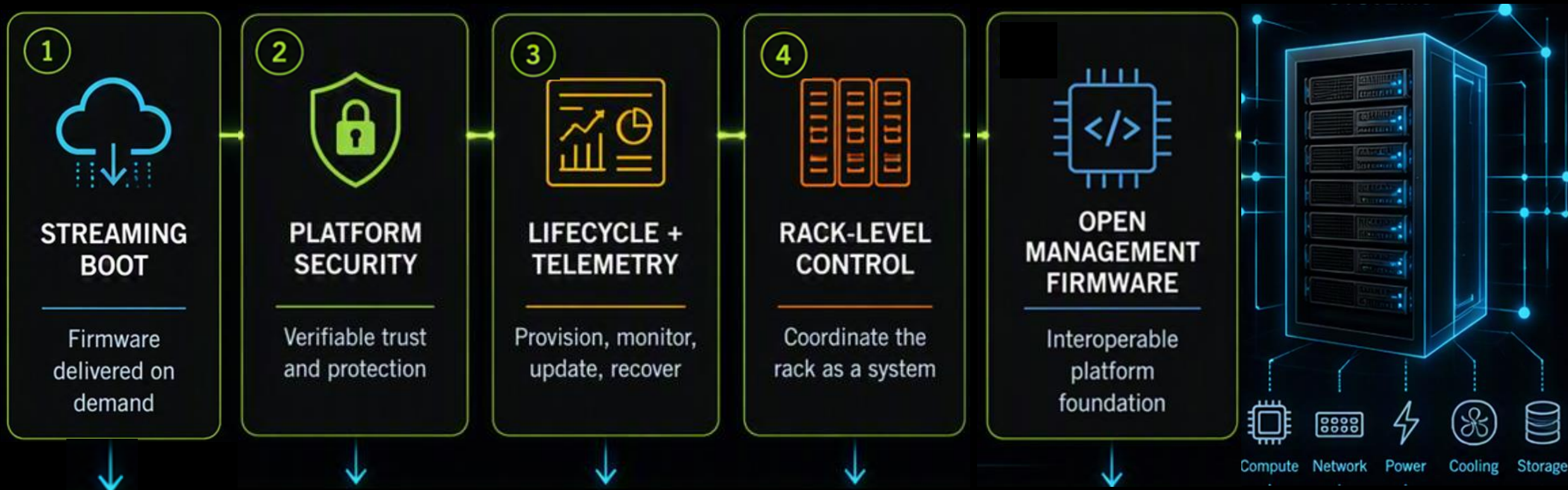
Effective control requires a common system-level contract connecting authority, topology, policy, state, commands, and evidence

FOUR GAPS

Bootup without a system level policy	Rack-level bring-up authority is unspecified
Authority without a contract	Attestation stops at the device boundary
Rack level unified control	Interfaces vary by ODM/OEM and generation
Telemetry w/o control evidence	Authorization, execution, outcome and recovery are not one verifiable record.

A Joint Approach to System Management

AI System



Need for interoperability and reducing integration complexity

Key Tenets For Use cases

Streaming Boot		Open Firmware	Operations Console	Fleet Dashboard	Policy Compliance		
Open Management Firmware		REST JSON	gRPC	Redfish	DC-SCM	PLDM	SNMP	Metrics
Platform Security		Device Identity	Secure Boot	Measure, Attest	Key Protection		
Lifecycle Telemetry		Compute Control	Dev. & Fabric Mng.	Sensors Telemetry	Policy Enforce	Firmware & Update	
System level Control		Policy Engine	Telemetry Aggr.	Power & Thermal	Security & Compliance	Lifecycle, Recovery	

Architecture: Capabilities for an Interoperable Solution

1. **Secure Firmware Distribution**
2. **Verified System Bring-Up**
3. **Attestation-Conditioned Control**
4. **Ownership, Revocation & Recovery**

The pieces exist. Now we need to connect them

Effective control is only achieved if the ecosystem aligns on the interfaces, trust model and validation approach.

Collaborate to Define the Control Architecture for the AI Era

- Define the boundary between trusted system bring-up and runtime management.
- Establish common contracts and effective control for system management, evidence and recovery
- Contribute operator requirements and production-platform validation.
- Build interoperable implementations and measure them against one specification.

OPEN INTERFACES • VERIFIABLE TRUST • REPLACEABLE IMPLEMENTATIONS